

Services Consultant Architect Practice Lab 2

IMPORTANT: The **approximate build time for this lab is 5 minutes**. Please be patient. If the lab does not build after 15 minutes, try refreshing the page and rebuilding it again. If it does not build on the second attempt, contact ellearn@splunk.com.

Lab Overview

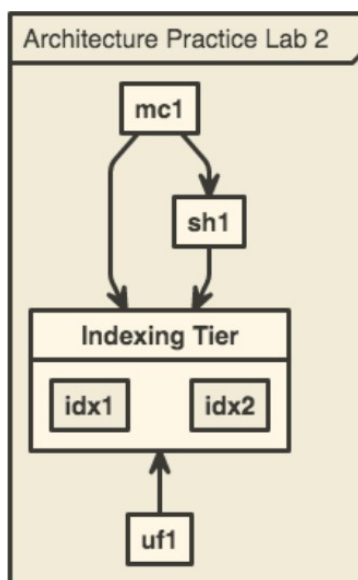
Maximum Estimated Duration – 4 Hours

This lab will focus on the configuration of a Splunk universal forwarder and app installation using a Splunk Deployment server.

You will configure a single Search Head, two Indexers and a Monitoring Console as in the previous lab and add an additional universal forwarder.

Your search head will be configured as the deployment server to manage the configuration and installation of apps on the universal forwarder.

Host Name	Role
ap2-xxxx-sh1	Search Head
ap2-xxxx-mc1	Monitoring Console
ap2-xxxx-idx1	Indexer
ap2-xxxx-idx2	Indexer
ap2-xxxx-uf1	Universal Forwarder



Configure your Distributed Search Infrastructure

Use the same steps to configure your infrastructure as in the first Practice lab; the Services Consultant Architecture Practice Lab 1.

As before - On each host excluding your new uf host:

1. Create A splunk User
2. Unarchive the Splunk Enterprise Version in /opt
3. Install Splunk as Root
4. Change the default Admin Password (Use the password provided in your Server list)
5. Run Splunk as a Non-Root User
6. Enable Splunk to Start on Boot as the Splunk User
7. Configure Your Indexers
8. Configure Distributed Search
9. Configure the Monitoring Console

Make sure to rename your base configuration templates with the correct prefix ap2_

Configure your Universal Forwarder Host

NOTE: The Splunk Enterprise installation package has been staged in the /opt folder on the lab systems.

1. Change to the /opt directory and issue the following command
`sudo tar -xvzf splunk_package_name.tgz`

Further information on the .tgz installation procedure can be found here:

http://docs.splunk.com/Documentation/Forwarder/latest/Forwarder/Installanixuniversalforwarder#Install_on_Linux

2. As the root user start the forwarder for the first time. (Use the password provided in your Server list).
<http://docs.splunk.com/Documentation/Splunk/latest/Installation/StartSplunkforthefirshme>
3. Enable Splunk to Start on Boot

Configure all of your nodes as a Deployment Client

Further information on the deployment server can be found here:

<http://docs.splunk.com/Documentation/Splunk/6.6.3/Updating/Aboutdeploymentserver>

1. Use the base configuration template "org_all_deployment_client" to configure your nodes as a deployment client of the search head. You will need to customize the deploymentclient.conf in the template app to communicate with your search head.
2. Make sure you rename the app to "ap2_all_deployment_client"
3. The following servers will need to be configured as deployment clients:
 - Monitoring Console
 - Indexers
 - Universal Forwarder

Configure Forwarding

Configure a base configuration app for data forwarding to the indexers.

1. Use the base configuration template "org_all_forwarder_outputs" to configure your UF to send logs to your indexers.
2. Rename the app from "org_all_forwarder_outputs" to "ap2_all_forwarder_outputs"
3. Update the server list under the [tcpout:primary_indexers] stanza in outputs.conf with your indexers
4. The same forwarder outputs app should be copied to both the apps and deployment-apps folders on your search head.

This will ensure that the search head internal logs are forwarded to the indexing tier in the same manner as the forwarder. Deployment server configuration will be completed later.

5. Restart splunk to ensure your configuration app is applied on all deployment client nodes.

Configure the Deployment Server ServerClass 1 - All Forwarders

1. Using the Forwarder Management UI configure your instances to send their logs to the indexing tier You should see all four clients connected to your deployment server.
2. Create a new serverclass titled "All_Forwarders"
3. Add a whitelist of "*uf1 , *mc1"
4. Add your updated ap2_all_forwarder_outputs app to the serverclass
5. Configure the serverclass to restart splunkd on update
6. From your search head run the following search to check your work index=_internal | stats count by host

You should see results from your all 5 of your instances

7. Finally configure forwarder monitoring in your Monitoring Console

Configure and deploy the Splunk Add-on for Unix and Linux

1. Go to Splunkbase and download the Splunk Add-on for Unix and Linux
2. Put the app into both the apps and deployment-apps folders on your search head.
3. Enable the following inputs in the Linux Add-on

```
[script:///bin/vmstat.sh]
[script:///bin/iostat.sh]
[script:///bin/ps.sh]
[script:///bin/top.sh]
[script:///bin/netstat.sh]
[script:///bin/bandwidth.sh]
[script:///bin/protocol.sh]
[script:///bin/openPorts.sh]
[script:///bin/time.sh]
[script:///bin/lsof.sh]
[script:///bin/df.sh]
[script:///bin/who.sh]
[script:///bin/usersWithLoginPrivs.sh]
[script:///bin/lastlog.sh]
[script:///bin/interfaces.sh]
[script:///bin/cpu.sh]
[script:///bin/rlog.sh]
[script:///bin/package.sh]
[script:///bin/hardware.sh]
[monitor:///var/log]
```

4. To enable the inputs, create a local inputs.conf file in the Add on.

Your inputs.conf file only needs the appropriate stanzas and a setting of disabled=false

5. Add the following to your local/inputs.conf in the TA. [default] index=os
6. Use the org_all_indexes base configuration template to create an os index.
7. Rename the app ap2_all_indexes.

Create/Configure ServerClass 2 - All Linux Hosts

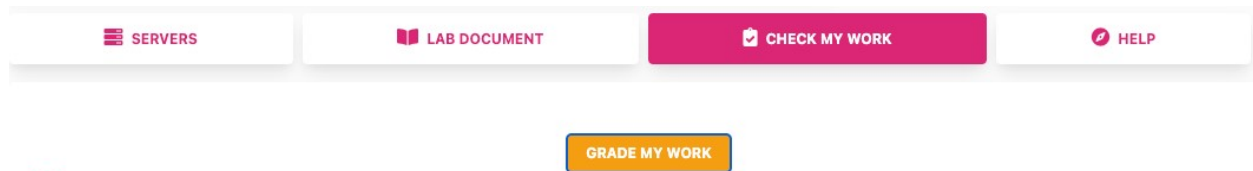
In this section you will use the Forwarder Management UI to configure your instances to send their logs to the indexing tier

1. Create a new serverclass titled "All_Linux_Hosts".
2. Add the nix TA app to the serverclass.
3. Add the ap2_all_indexes app to the serverclass.
4. Configure the apps to restart splunkd on update.
5. Use a machinetypesfilter of "linux-x86_64" and a whitelist of "*".
6. Manually copy the add-on to the apps folder on your search head.
7. From your search head run the following search to check your work.
index=os | stats count by host

You should see results from your all 5 of your instances.

Grading Your Work

Grading of the lab is done through the same interface where your student instances are displayed.



Grading of the labs is an intensive process involving many calls to the instances. This may take several minutes to complete. Please be patient. If the grading does not complete in 5 minutes, please reach out to ellearn@splunk.com for assistance.

For best results, make sure you complete all of the lab steps before grading the lab.

1. Click the Check My Work Tab.
2. Click the Grade My Work button.

After a few minutes, your results will be displayed.

You must pass all graded items to receive credit for the lab. Resolve any failed grading items and re-issue the grading check.



If you pass all of the lab grading checks, the Learning Management System (LMS) will terminate the browser window. Log back into the LMS to view your completed status for the lab.